



Ignite IT Performance™

Waits and Queues and You

Thomas LaRock
Senior DBA, Confio Software



Who Am I?

CONFIO
SOFTWARE
Ignite IT Performance™

Microsoft
CERTIFIED
Database Administrator



ignite8
CONFIO®



follow me
@SQLRockstar



<http://thomaslarock.com>

Learning to Drive



- Gas gauge



- Gas gauge
- Oil light





Learning to Drive

- Gas gauge
- Oil light
- Odd engine noises





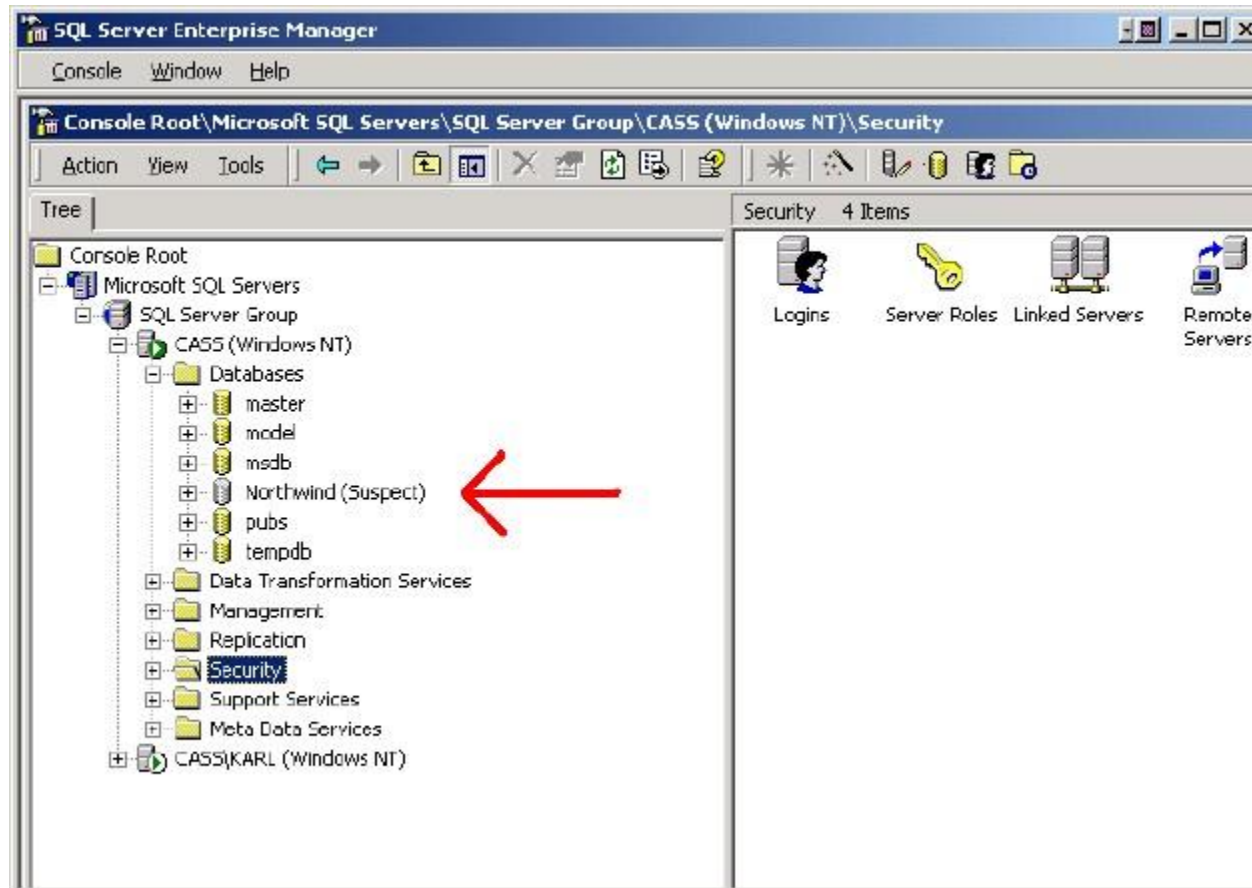
Learning to Drive

- Gas gauge
- Oil light
- Odd engine noises
- What about DBA?





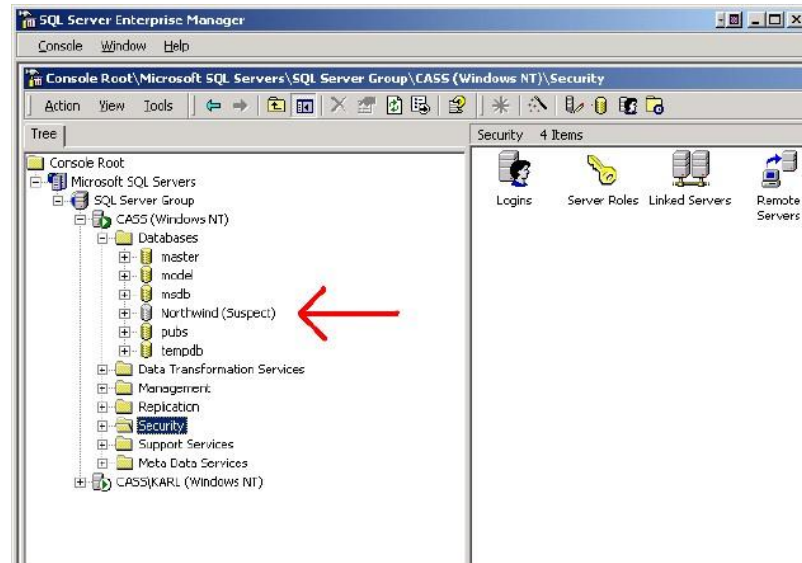
Learning to DBA





Learning to DBA

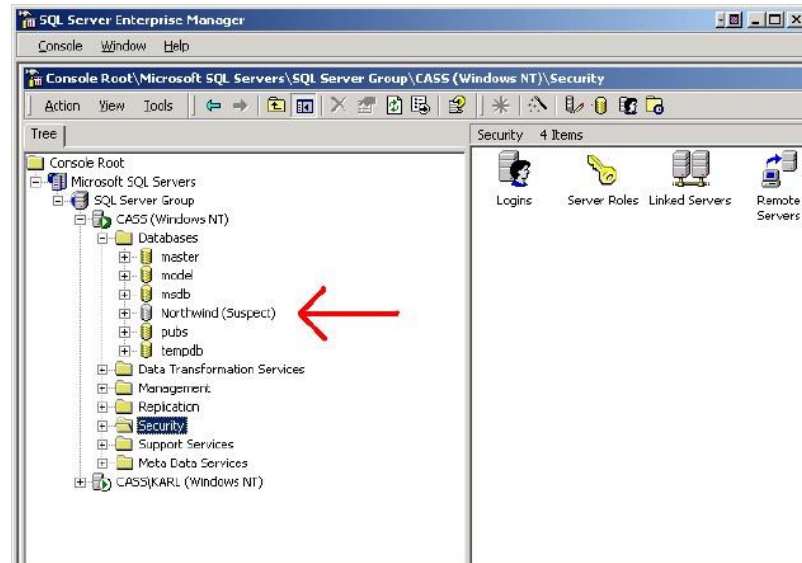
- Red/green icons





Learning to DBA

- Red/green icons
- Phone ringing



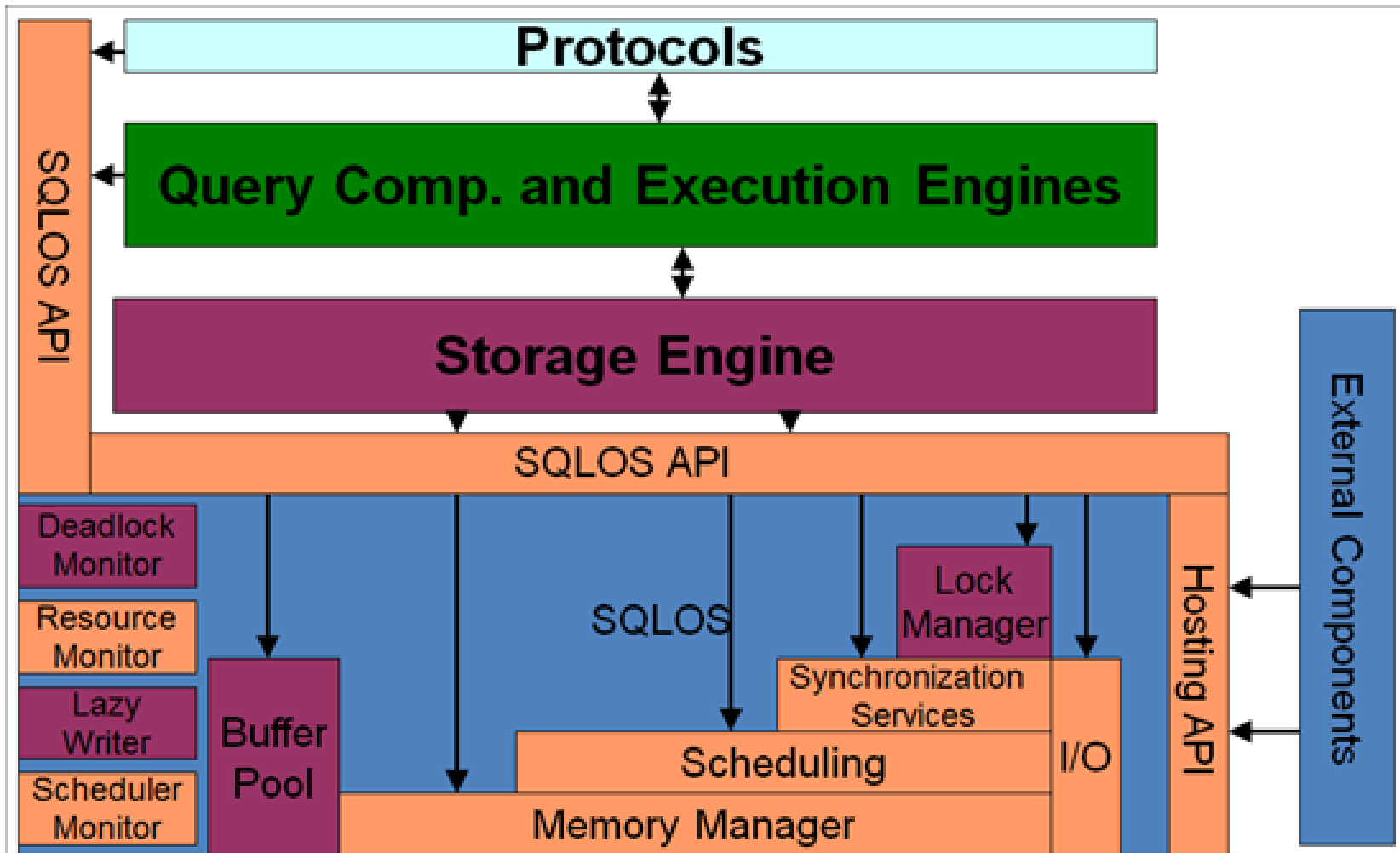


SQL Server is a Black Box

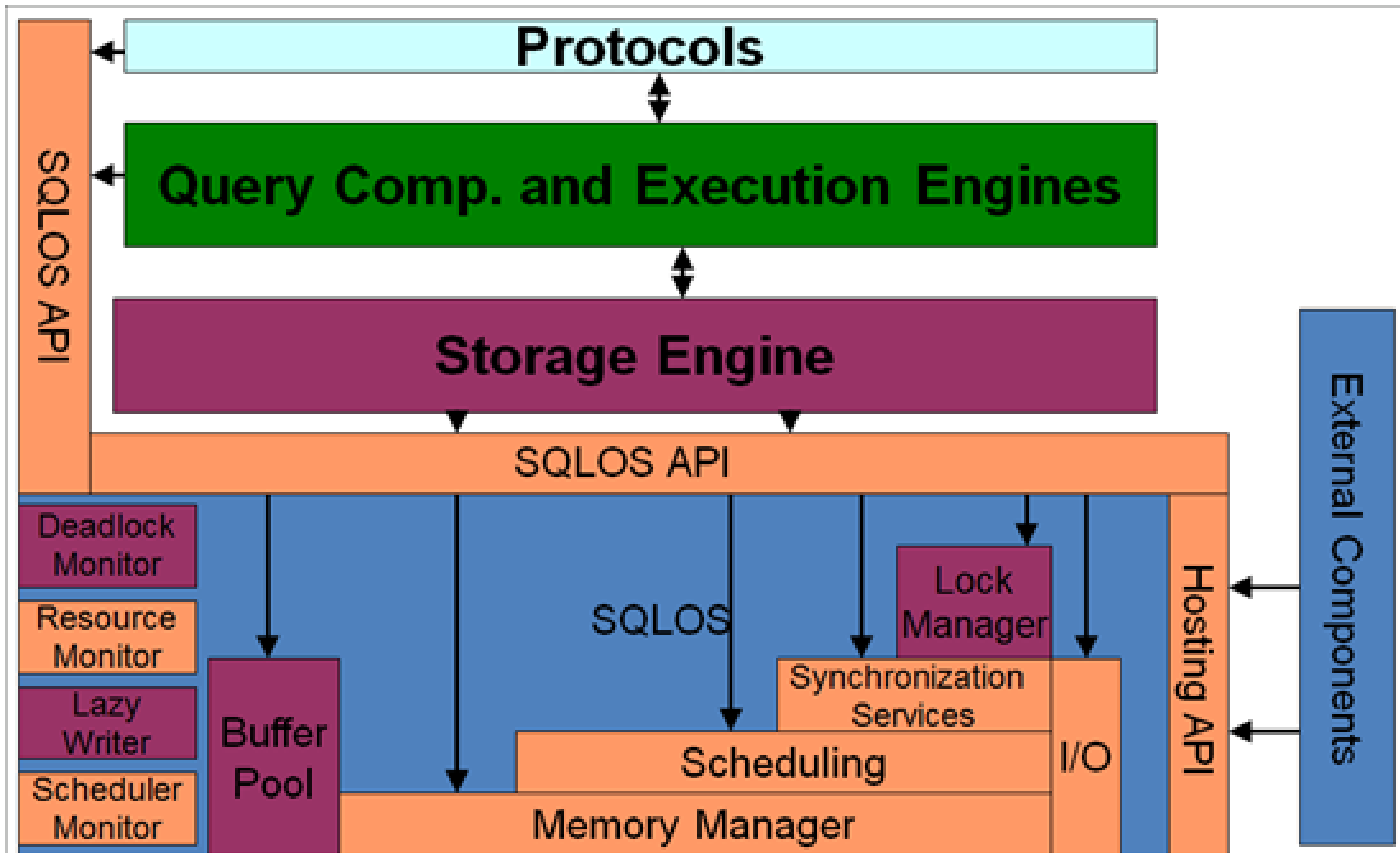


- Oil changes
- Tune ups
- Alignments
- Brakes





[illegible]



- Describe Execution Model



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- Use some DMVs



Agenda

- Describe Execution Model
- Use some DMVs
- Show why we measure waits (use case)



Execution Model Example



Your Tax Dollars at Work





Your Tax Dollars at Work





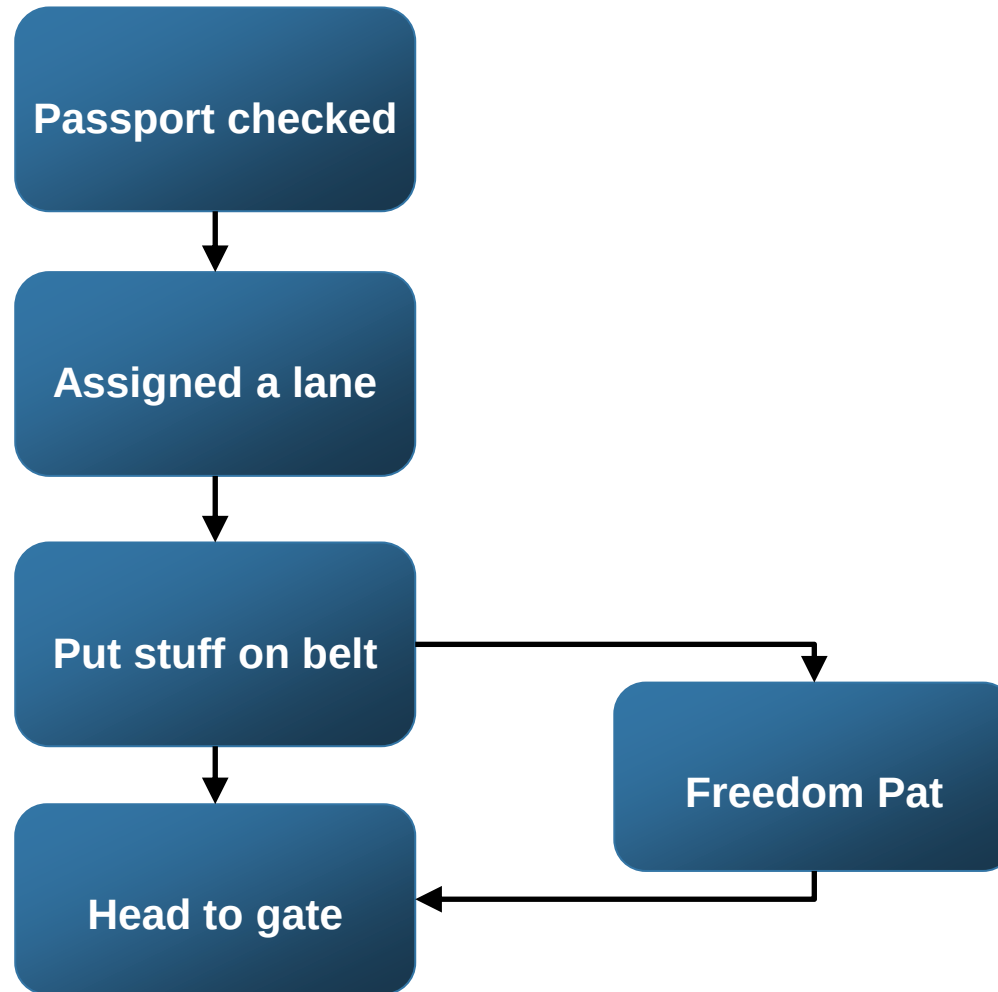






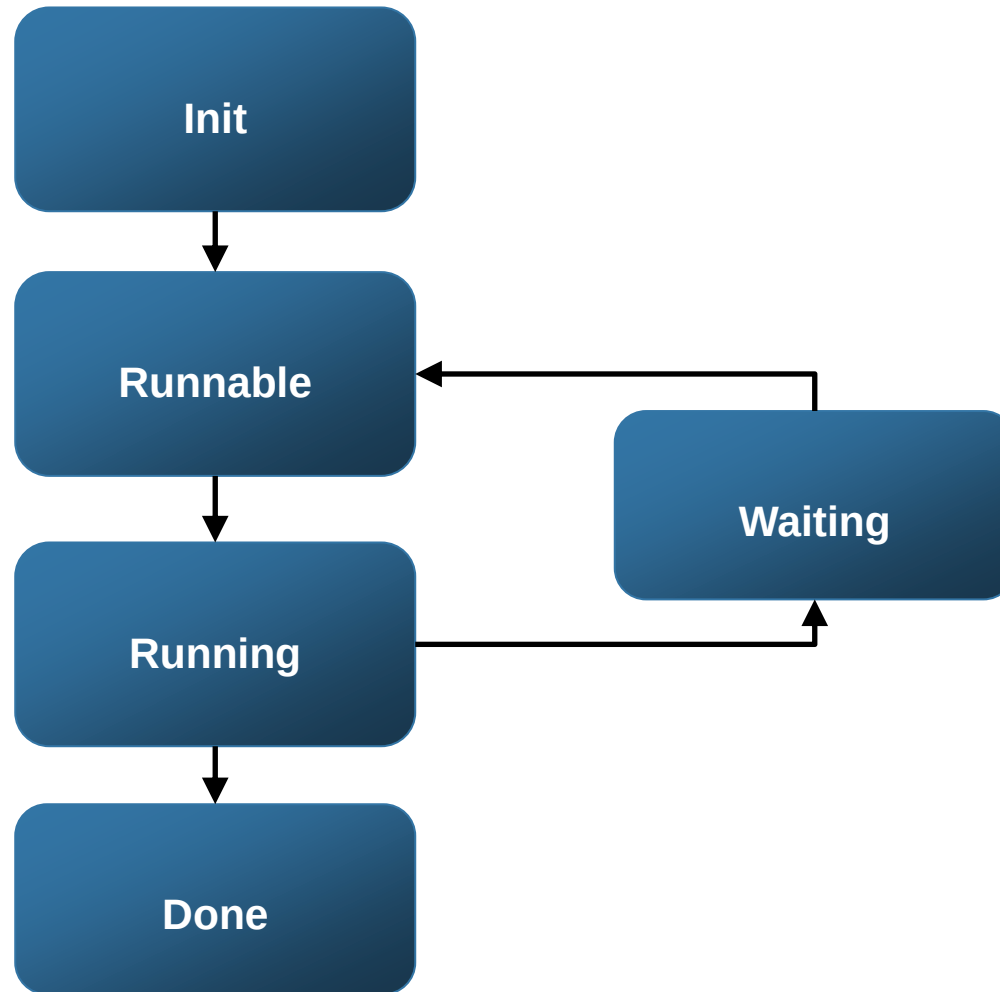


Execution Model (TSA)





Execution Model (SQLoS)





Execution Model (SQLOS)

CPU 1

SPID 77 – Running

CPU 1 Queue

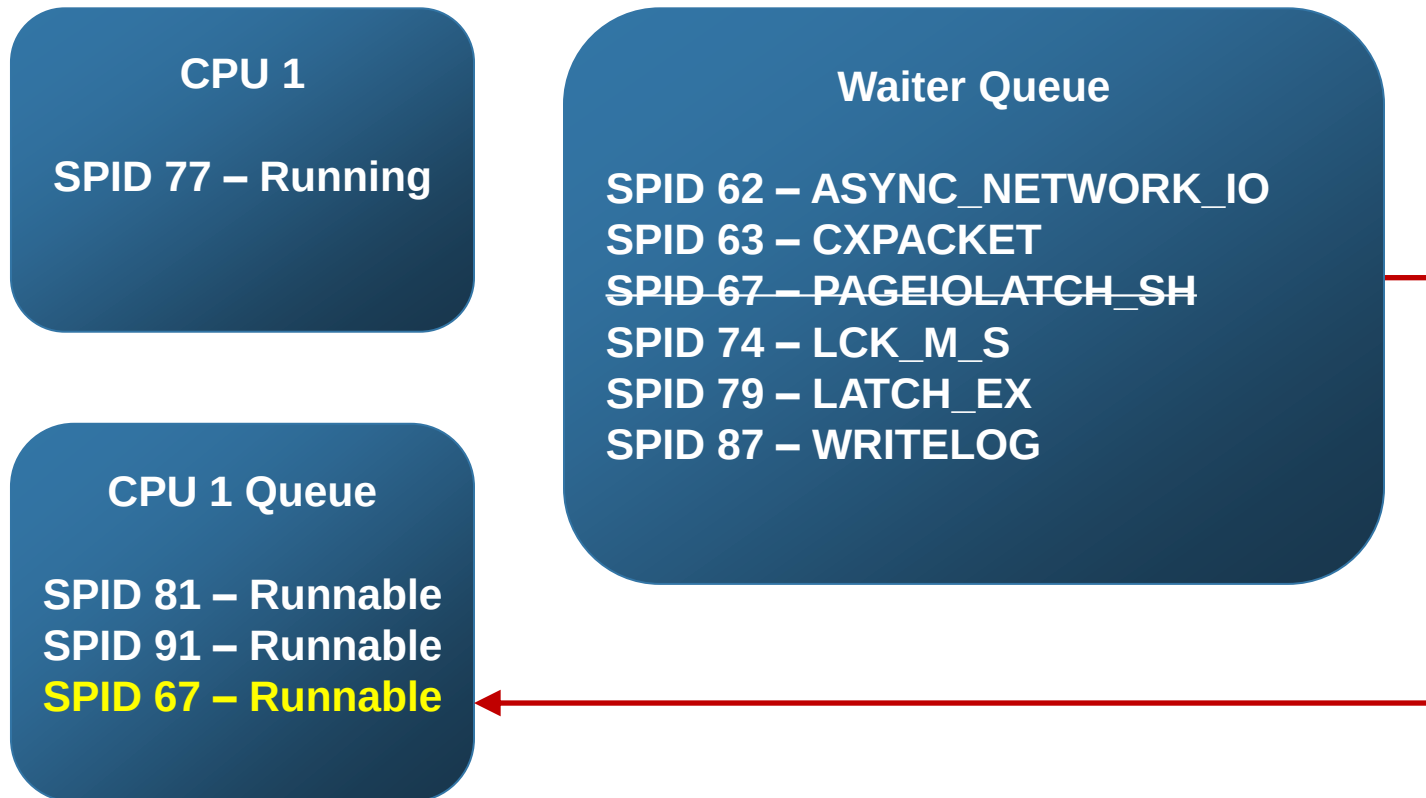
SPID 81 – Runnable
SPID 91 – Runnable

Waiter Queue

SPID 62 – ASYNC_NETWORK_IO
SPID 63 – CXPACKET
SPID 67 – PAGEIOLATCH_SH
SPID 74 – LCK_M_S
SPID 79 – LATCH_EX
SPID 87 – WRITELOG

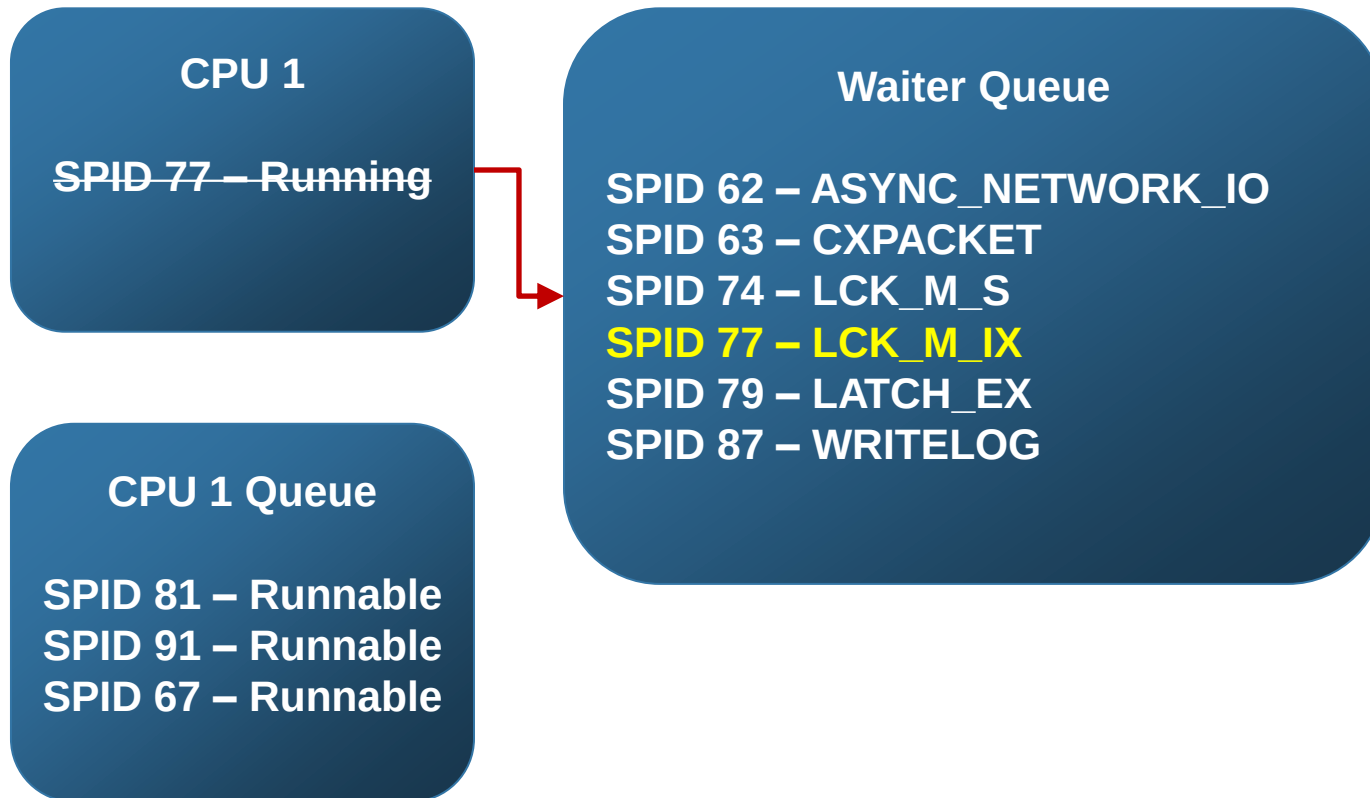


Execution Model (SQLOS)



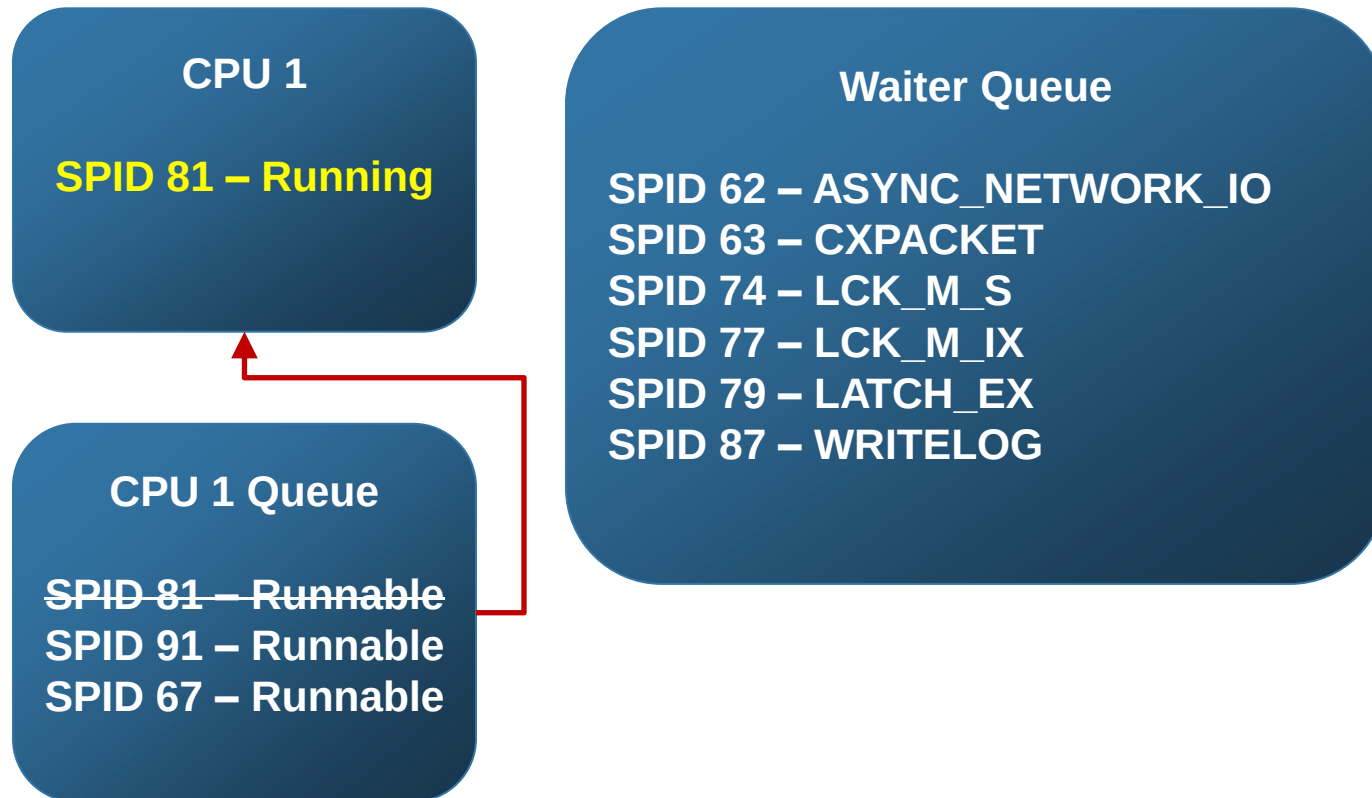


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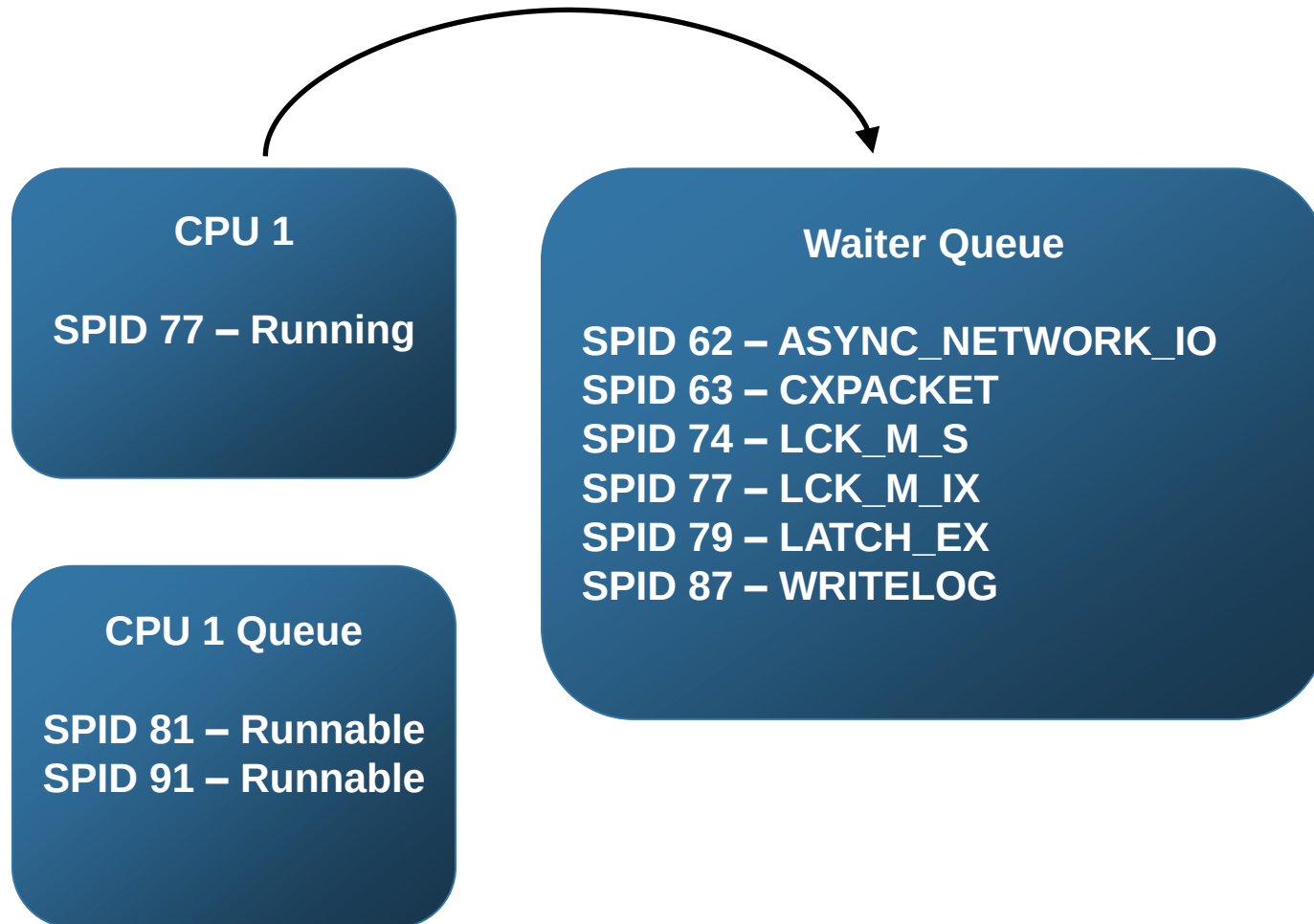


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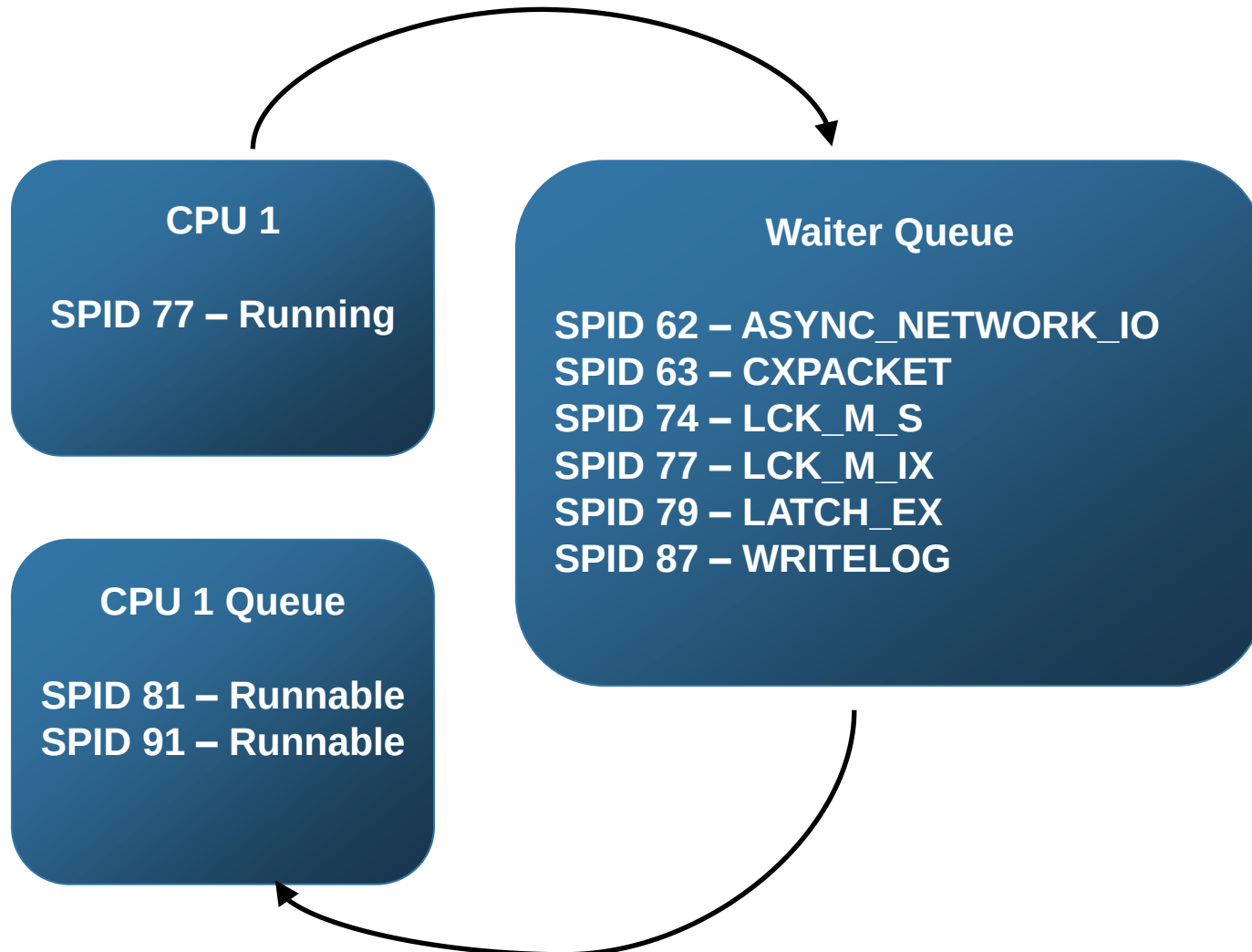


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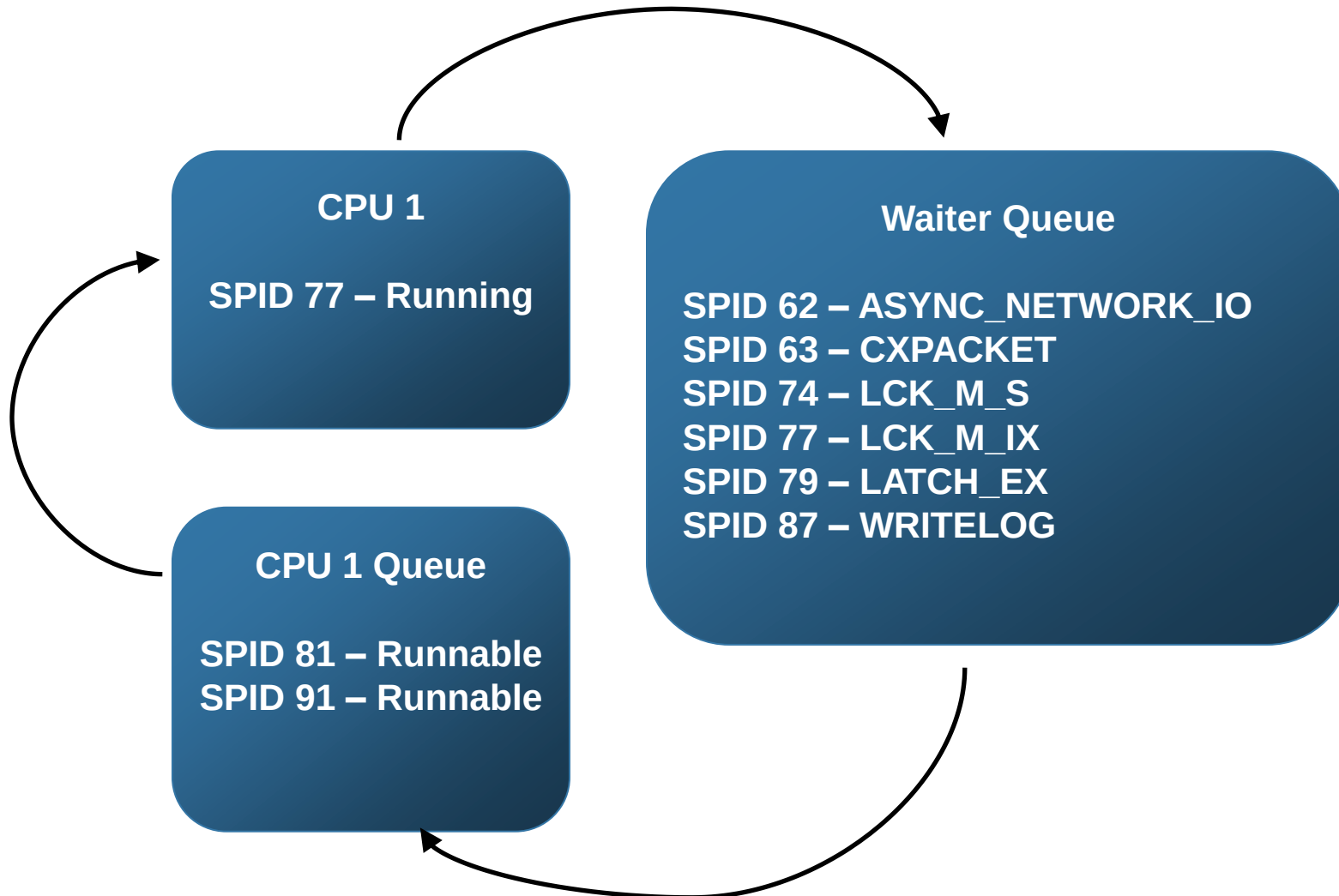


Execution Model (SQLOS)



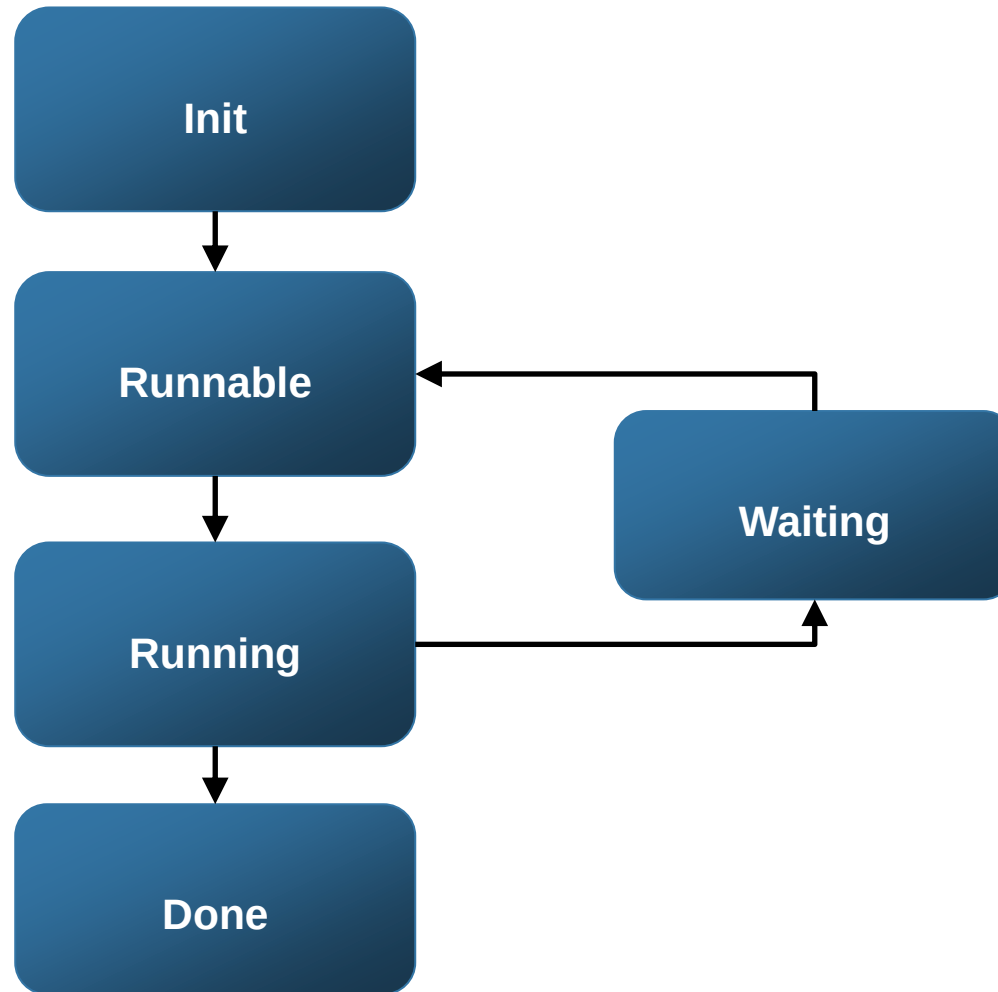


Execution Model (SQLLOS)



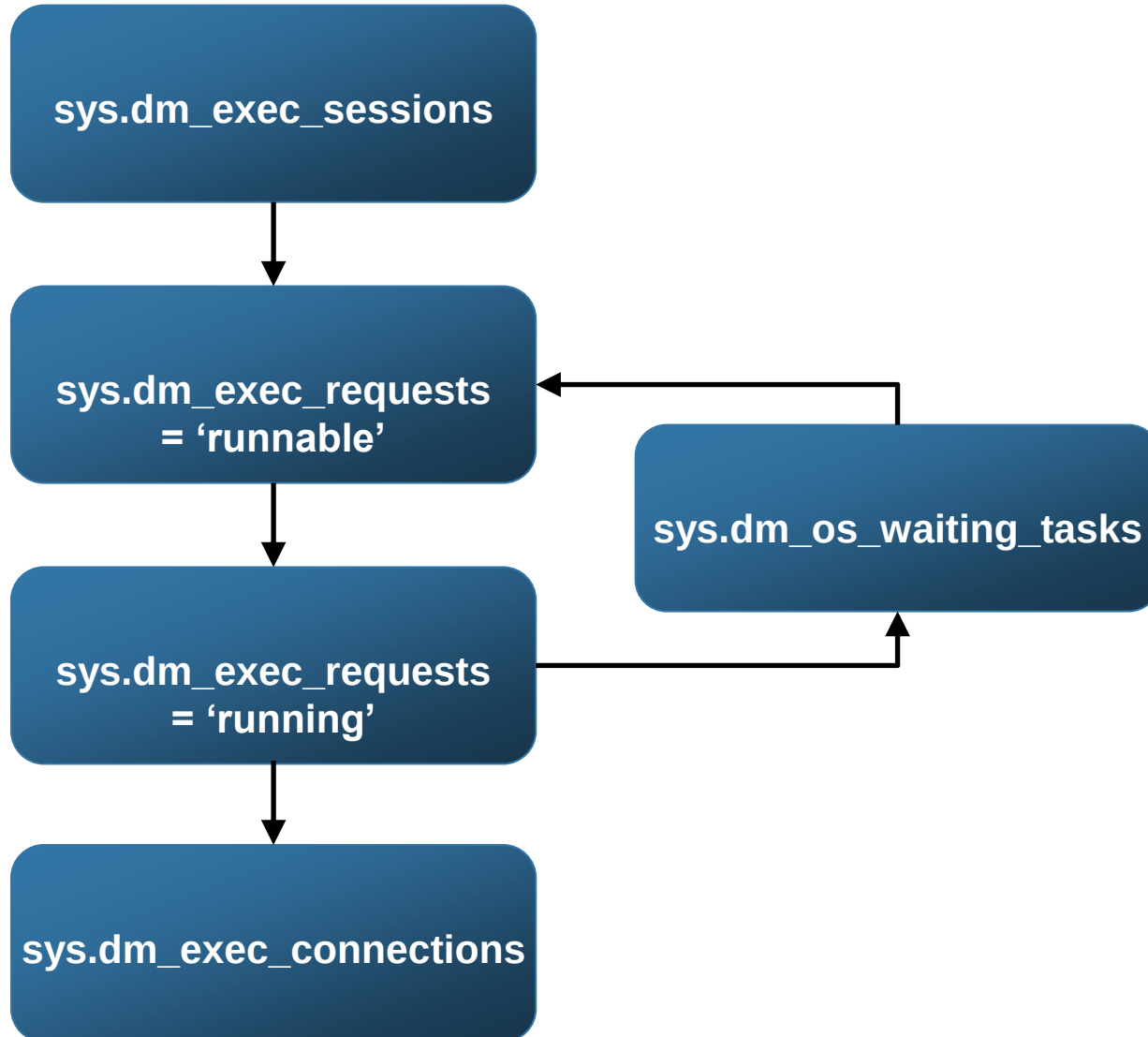


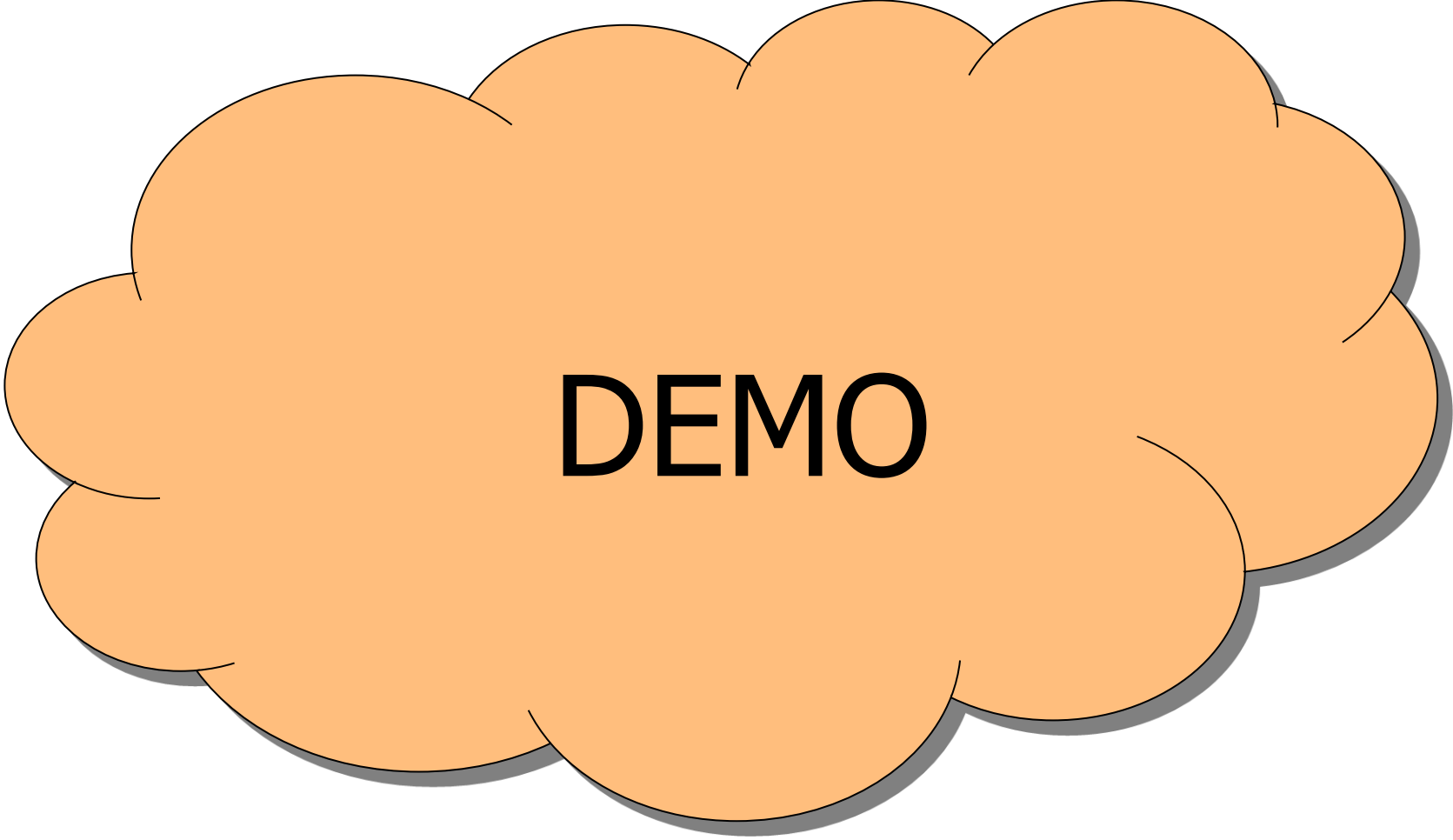
Execution Model (DMVs)





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DEMO



Sample Wait Types

1.

2.

3.

4.

5.



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2.

3.

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2.

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5. LCK_M_S, LCK_M_U, LCK_M_X...

- Waiting to acquire locks



1.

2.

3.

4. PAGEIOLATCH_SH, PAGEIOLATCH_EX...

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1.

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- Physical disk

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1.

2.

3. WRITELOG

4. PAGEIOLATCH_SH, PAGEIOLATCH_EX...

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- Waiting for a log flush to complete

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1. ASYNC_NETWORK_IO, NETWORKIO
 - Waiting on the network
2. CXPACKET
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Why Measure Waits?

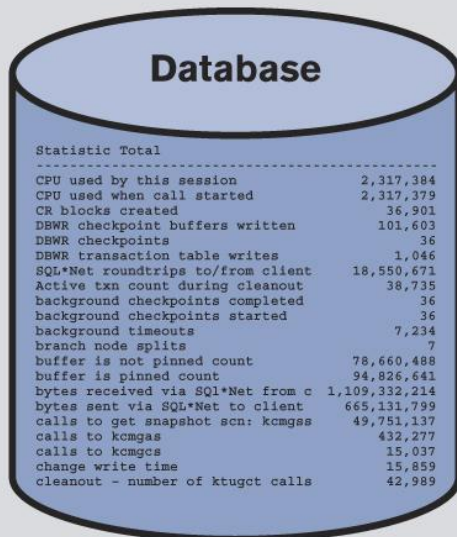
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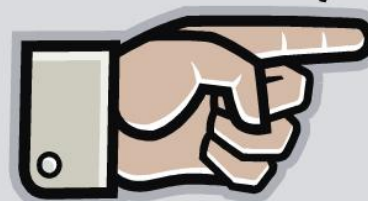
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Conventional Tools Measure Database Health



Statistic Total	
CPU used by this session	2,317,384
CPU used when call started	2,317,379
CR blocks created	36,901
DBWR checkpoint buffers written	101,603
DBWR checkpoints	36
DBWR transaction table writes	1,046
SQL*Net roundtrips to/from client	18,550,671
Active txn count during cleanup	38,735
background checkpoints completed	36
background checkpoints started	36
background timeouts	7,234
branch node splits	7
buffer is not pinned count	78,660,488
buffer is pinned count	94,826,641
bytes received via SQL*Net from c	1,109,332,214
bytes sent via SQL*Net to client	665,131,799
calls to get snapshot scn: kcmgss	49,751,137
calls to kcmgas	432,277
calls to kcmgcs	15,037
change write time	15,859
cleanup - number of ktugct calls	42,989

It's your
Code!



DBA

It's your
Database!



**Developer or
Vendor**

Unclear view of performance leads to finger pointing.



Why Wait Time Analysis Rocks

- Allows you to focus on *the* problem
 - Four possible resource bottlenecks



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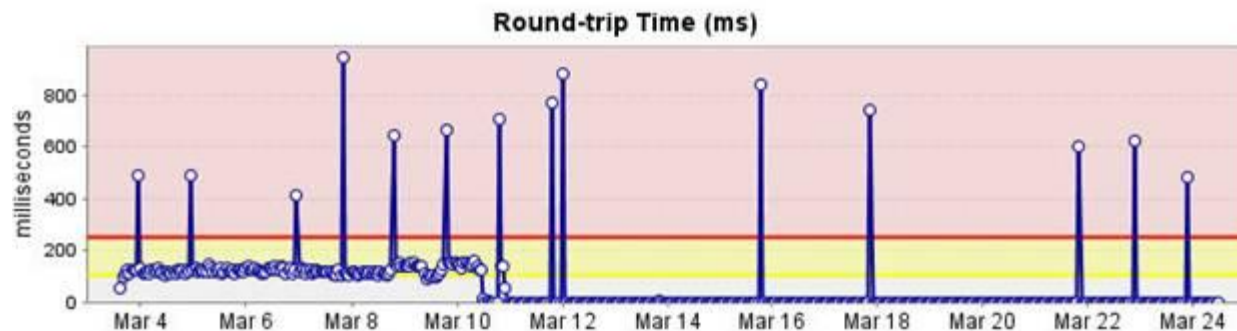
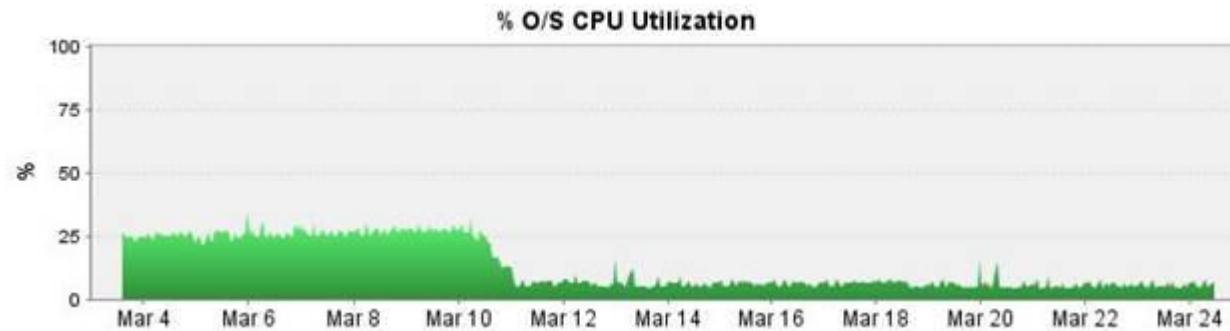
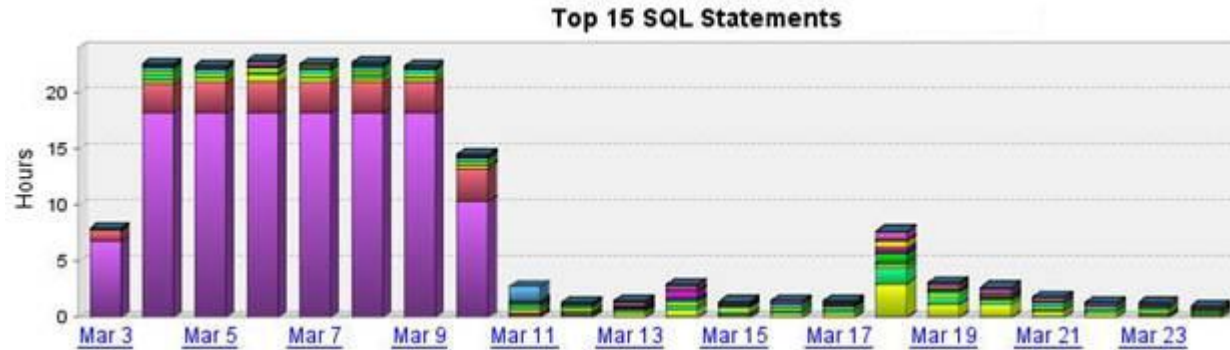


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 - Four possible resource bottlenecks
- Waits help you to answer the BIG question:
 - “Is this a problem?”
- Helps to prioritize
 - Fix the ones you can fix now
- Does not rely on Health Stats alone
 - Perfmon counters only?



Use Case (or, My Story)





How To Measure?

- 3rd party tools
- Native tools
 - DMVs
 - Perfmon
 - xEvents
 - MDW/UCP

- Know difference between “wait” and “queue”
 - Wait – Any time a session is waiting
 - Queue – A measure of system resource and utilization (think perfmon)

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- What does all that mean to the DBA?



Operating With Precision



THANK YOU CLEVELAND!
NEXT STOP: ST. LOUIS!

Questions?





For More Information

- <http://sqlcat.com/whitepapers/archive/2007/11/19/sql-server-2005-waits-and-queues.aspx>
- <http://technet.microsoft.com/en-us/sqlserver/gg508897.aspx>
- <http://sqlskills.com/BLOGS/PAUL/post/Wait-statistics-or-please-tell-me-where-it-hurts.aspx>
- <https://sqlserverperformance.wordpress.com/>
- <http://speakerrate.com/speakers/3000-sqlrockstar>